ICT220: Data Structure and Algorithm (C++, C# or JAVA)

Rational

Object-Oriented Programming and Algorithms is designed to equip students with fundamental programming principles and algorithmic problem-solving skills. By comprehensively exploring concepts of Object-Oriented Programming (OOP), data structures, and algorithm analysis, students will gain a strong foundation in software development. Understanding the principles of OOP, including inheritance, encapsulation, and polymorphism, enables them to design efficient and maintainable code. By delving into algorithm design and analysis techniques, empowering students to create optimized solutions to complex problems. These skills are invaluable in today's technology-driven world, as they form the basis for developing software applications, solving real-world challenges, and pursuing careers in software development, data science, and more.

Learning outcomes

The main aim of this course is to provide students with a strong foundation in Object-Oriented Programming (OOP), algorithm analysis, and data structures. This knowledge will enable them to design efficient algorithms, implement data structures, and tackle complex computational problems. It will enhance students' problem-solving skills and prepare them for diverse software development and computer science challenges.

Main Objective

By the end of the course-tenure the students should to:

- Define and explain key Object-Oriented Programming (OOP) concepts and terminologies.
- Differentiate between various access modifiers and their applications in OOP.
- Describe the characteristics and components of objects in OOP.
- Apply the principles of inheritance, polymorphism, and encapsulation in program development.
- Understand and work with arrays and string-based sequences, including multi-dimensional arrays and string manipulation.
- Analyze and evaluate algorithms using mathematical techniques for time and space complexity.

- Implement and manipulate data structures like lists, stacks, and queues.
- Demonstrate proficiency in tree and graph structures, including binary trees and graphs traversal.
- Implement sorting algorithms, such as bubble sort, insertion sort, and quicksort.
- Apply various algorithm design techniques, including greedy algorithms, divide and conquer, dynamic programming, and backtracking.

Course content

Unit 1: Concepts of Object Oriented Programming

- Definitions and terminologies
- Types of Access Modifier
- Characteristics of an object
- Concepts of inheritance, polymorphism and encapsulation

Unit 2: Arrays and String based sequences

- Single dimensional array
- Multi-dimensional array
- Passing an array to a function
- String concepts
- Insertion, deletion, and updating array elements
- Insertion, deletion, and updating string elements
- Pointers

Unit 3: Algorithm Analysis

- How to write an algorithm
- Mathematical Background for time and space complexity
- What to Analyze
- Best, Worst, and Average Case Complexity
- Running-Time Calculations

Unit 4: Lists, Stacks, and Queues

- Types of lists
- Operations on lists
- Implementation of stacks and queues
- Operations on stacks and queues

Unit 5: Trees and Graphs

- Basic Terminology
- Binary Trees
- Heaps
- AVL Trees
- Directed and Undirected Graphs
- Depth-First Search
- Breadth-First Search

Unit 6: Sorting Techniques

- Bubble sort
- Insertion Sort
- Merge sort
- Quicksort

Unit 7: Algorithm Design Techniques

- Greedy Algorithms
- Divide and Conquer
- Dynamic Programming
- Backtracking Algorithms

Teaching and Learning Methods

Lectures:90Hrs Tutorials:60Hrs, Labs:60Hrs, Assessments:30Hrs, NATIONAL HRS: 240

Assessment

- Continuous Assessment 40%
  - 20% for hands-on assessments
  - 20% for non-hands-on assessments
- Final Exams 60%
  - 20% for hands-on assessments
  - 30% for theory assessments

Recommended Readings

- Christopher Fox (2018), Go Data Structures and Algorithms, Christopher Fox & bookboon.com
- Michael T. Goodrich and Roberto Tamassia (2004), Data Structures and Algorithms in Java, John Wiley & Sons, Inc.
- Julian Templeman (2013), Microsoft Visual C++/CLI Step by Step, O'Reilly Media, Inc.